

Experiment 8

Test Case 1: Design a Two conductor transmission line with single ground plane and investigate the impact of the width of the rectangular conductor on Characteristic Impedance and Propagation Velocity .Thickness to be varied between 10 to 20 mm.

AIM: To design a Two conductor transmission line with single ground plane and investigate the impact of the width of the rectangular conductor on Characteristic Impedance and Propagation Velocity .Thickness to be varied between 10 to 20 mm.

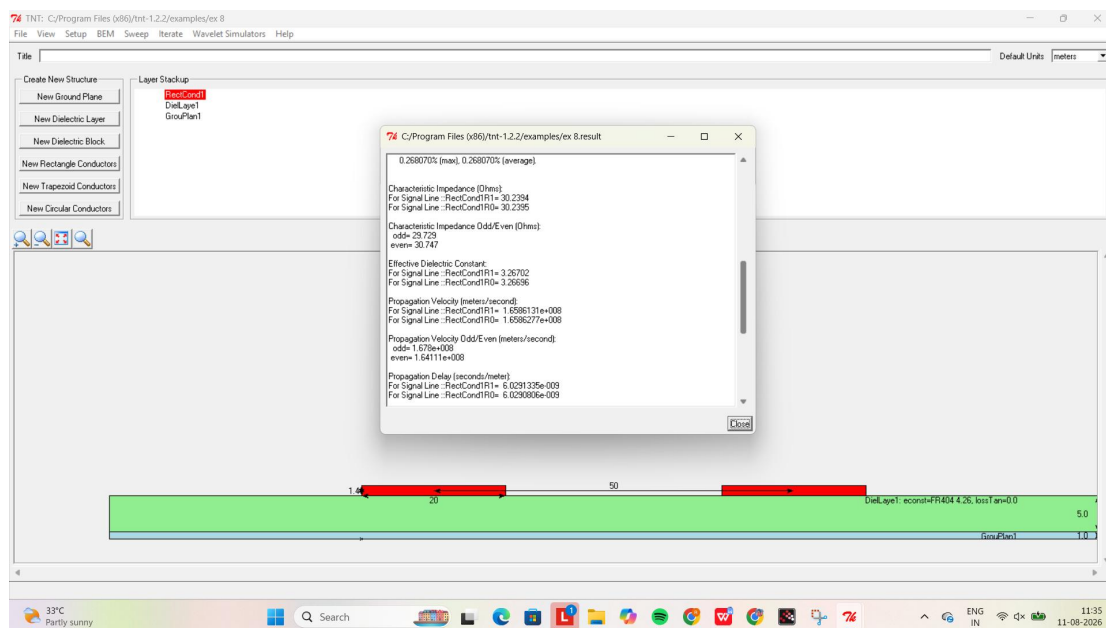


Table-

Width of Conductor (mm)	Characteristic Impedance (ohms)	Propagation Velocity (m/s)
20	30.239	1.65e+008
18	32.55	1.67e+008

Test Case 2: Examine the effect of Characteristic Impedance and Propagation Velocity by varying the height of the rectangular conductor . Height to be varied between 1 to 1.4 mm.

AIM: To examine the effect of Characteristic Impedance and Propagation Velocity by varying the height of the rectangular conductor .Height to be varied between 1 to 1.4 mm.

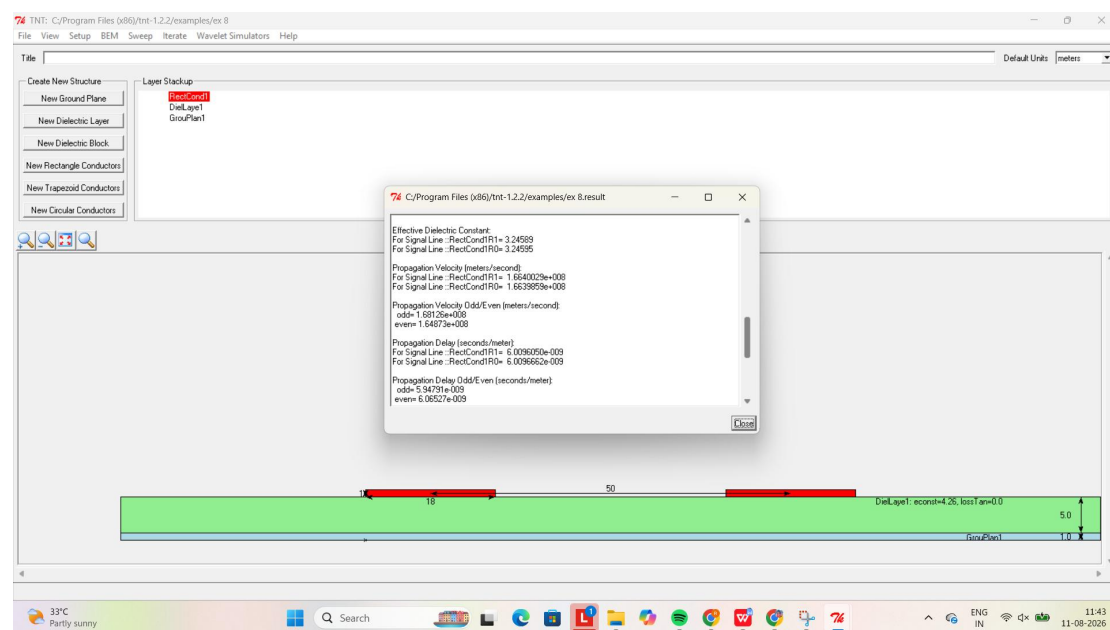


Table-

Height of Conductor	Characteristic Impedance (ohms)	Propagation Velocity (m/s)
1	32.80	1.664e+008
18	32.55	1.67e+008